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# Tutorial - 7

## (KNN)

K - Nearest Neighbour

**Training dataset:**

| Action scence | Romance scenes | Happy Ending | Genre |
|---|---|---|---|
| 15 | 2 | 0 | A |
| 1 | 12 | 1 | R |
| 18 | 1 | 0 | A |
| 2 | 14 | 1 | R |
| 14 | 3 | 1 | A |

**Test Dataset (with Actual Labels):**

| Test ID | Action scenes | Romance scene | Happy Ending | Actual Genre |
|---|---|---|---|---|
| T1 | 10 | 5 | 1 | A |

**comparing** (10, 5, 1)

(15, 2, 0)

(1, 12, 1)

(18, 1, 0)

(2, 14, 1)

(14, 3, 1)

$(10,5,1)\ (15,2,0) = \sqrt{(10-15)^2+(5-2)^2+(1-0)^2} = \underline{5.917}$ - A

$(10,5,1)\ (1,12,1) = \sqrt{(10-1)^2+(5-12)^2+(1-1)^2} = \underline{11.402}$ R

$(10,5,1)\ (18,1,0) = \sqrt{(10-18)^2+(5-1)^2+(1-0)^2} = \underline{9}$ - A

$(10,5,1)\ (2,14,1) = \sqrt{(10-2)^2+(5-14)^2+(1-1)^2} = \underline{12.042}$ R

$(10,5,1)\ (14,3,1) = \sqrt{(10-14)^2+(5-3)^2+(1-1)^2} = \underline{4.473}$ - A

nearest distances: 5.92 — A. (k = 3)

9.0 — A.

4.74 — A

since weight

$$= \frac{1}{d} = \frac{1}{5.92} = 0.168$$

$$= \frac{1}{9.0} = 0.11$$

$$= \frac{1}{4.74} = 0.2109$$

$w_1 = \frac{1}{5.92} = 0.168$

$w_2 = \frac{1}{11.40} = 0.08$

$w_3 = \frac{1}{9.0} = 0.11$

$w_4 = \frac{1}{12.04} = 0.08$

$w_5 = \frac{1}{4.47} = 0.22$

weights of A

$= w_1 + w_3 + w_5$

$= 0.52$

weights of R

$= w_2 + w_4$

$= 0.16$

$0.52 < 0.16$

$A > R$

so, we have got the 100% accurate result for training to testing data.

It is for testing, the testing data

Ex: If we have 4-values, 2 are R and 2 are A then 50% accuracy so, we need to calculate the weights by this whose value is greater than the other is treated as important / predicted value

# Regression (KNN)

| Age ($x_1$) | Previous value ($x_2$) in lakhs | Car Price ($y$) in Lakhs |
|---|---|---|
| 2 | 3 | 2 |
| 3 | 4 | 3 |
| 5 | 1 | 0.5 |

Test :- $(4, 2, y) \rightarrow 1.75$
$x_1 \quad x_2 \quad 5$

$(4,2) \; (2,3) = \sqrt{(4-2)^2 + (2-3)^2} = 2.24$

$(4,2) \; (3,4) = \sqrt{(4-3)^2 + (2-4)^2} = \underline{2.24} \rightarrow 3$ car price

$(4,2) \; (5,1) = \sqrt{(4-5)^2 + (2-1)^2} = \underline{1.41} \rightarrow 0.5$

$\boxed{k = 2}$

$$\hat{y} = \frac{1}{k} \sum_{i=1}^{k} y_i = \frac{3 + 0.5}{2} = 1.75$$

$(4, 2, 5)$ $\quad y = 5$

Actual $\quad \hat{y} = 1.75$